

B. Tech Degree VI Semester Examination April 2011**ME 601 INSTRUMENTATION AND CONTROL SYSTEMS**
(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

PART – A
(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Briefly explain the elements of a generalized measurement systems.
 (b) What do you mean by a zero order instrument?
 (c) What do you mean by gauge factor? Obtain an expression for gauge factor.
 (d) Explain the working principle of a sound level meter.
 (e) Define and explain transfer functions.
 (f) Explain the concept of steady state error and static error constants.
 (g) Define BIBO stability. What is the requirement for BIBO stability?
 (h) Explain the working principle of DC Tachogenerator.

PART – B

(4 x 15 = 60)

- II. (a) What do you mean by order of an instrument in the mathematical modeling? Define time constant and static sensitivity of a first order instrument. (6)
 (b) Derive the equations for time response of a first order system when subjected to
 (i) unit step input (ii) unit ramp input
 Draw response curves and find the steady error in each case. (9)

OR

- III. (a) Explain the three categories of error in measuring systems. (6)
 (b) Explain various systematic errors giving suitable examples. (9)
- IV. (a) What do you mean by un bonded and bonded strain gauges? (5)
 (b) Explain the methods of calibration of strain gauges. (10)

OR

- V. (a) Write notes on :
 (i) hydraulic load cells (ii) Froude hydraulic dynamometer. (7)
 (b) Briefly explain ORSAT's apparatus with neat sketches. (8)

- VI. For a unity feed back control system the open loop transfer function

$$G(s) = \frac{10(s+2)}{s^2(s+1)} \text{ Find}$$

- (i) the position, velocity and acceleration error constants
 (ii) the steady state error when the input is $R(s) = \frac{3}{s} - \frac{2}{s^2} + \frac{1}{3s^3}$. (15)

OR

- VII. Find the steady state error when the input is

- (i) unit step signal
 (ii) unit ramp signal
 (iii) unit parabolic signal (15)

- VIII. (a) Briefly explain- Routh-Hurwitz criterion. (5)
 (b) Using Routh criterion, determine the stability of the system represented by the characteristic equation, $s^4 + 8s^3 + 18s^2 + 16s + 5 = 0$. Comment on the location of the roots of characteristic equation. (10)

OR

- IX. Sketch Bode Plot for the following transfer function and determine the system gain K for the gain cross over frequency to be 5 rad/sec.

$$G(s) = \frac{Ks^2}{(1+0.25s)(1+0.025s)}$$

(15)